MODULE 17 · DOMAIN 2 - ML MODEL AND FM DEVELOPMENT

Evaluating traditional models

Which metric, which threshold, and why accuracy is usually the wrong answer.

MODULE 17 of 43 DOMAIN 2 - Model Dev WEIGHT 24%

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Fraud is 0.2 percent of transactions. Predicting 'not fraud' every time scores 99.8 percent.

The confusion matrix

	MODEL SAID POSITIVE	MODEL SAID NEGATIVE
Actually positive	True positive (TP)	False negative (FN)
Actually negative	False positive (FP)	True negative (TN)

ON THE EXAM

Every classification metric is arithmetic on these four numbers. Learn the grid and the formulas follow.

False positives and false negatives are not equivalent

- **False positive:** a false alarm. You acted when you should not have.
- **False negative:** a miss. You did nothing when you should have acted.
- **Ask what each one costs** in the actual business. The costs are almost never equal.
- A missed cancer diagnosis and an unnecessary follow-up scan are not the same size of mistake.
- A blocked legitimate payment and an approved fraudulent one are not either.

> ON THE EXAM
>
> This is the question that chooses the metric. Everything after this slide follows from which error you can least afford.

Precision: when you say positive, are you right?

- **Precision = TP / (TP + FP).** Of everything flagged, how much was genuinely positive?
- **High precision means few false alarms.** It ignores what you missed entirely.
- Optimise it when acting on a positive is **expensive, disruptive or irreversible**.
- Spam filtering: a legitimate email in the spam folder is worse than spam in the inbox.
- Also called positive predictive value.

> ON THE EXAM
>
> Precision answers: can I trust this alert? It says nothing about how many alerts never fired.

Recall: of everything positive, how much did you find?

- **Recall = TP / (TP + FN).** Of everything actually positive, how much did you catch?
- **High recall means few misses.** It ignores how many false alarms you raised.
- Optimise it when **missing a positive is dangerous or costly**.
- Disease screening, fraud detection, safety inspection, equipment failure.
- Also called sensitivity, or the true positive rate.

> ON THE EXAM
>
> Recall answers: did I find them all? It says nothing about how much noise I generated doing it.

Precision or recall?

FAVOUR PRECISION WHEN

A false alarm is expensive

- Acting on a positive costs real money or trust.
- A human has to review every flag.
- Blocking a legitimate case causes harm.
- Spam, content moderation, marketing spend.

FAVOUR RECALL WHEN

A miss is expensive

- Missing a positive causes serious harm.
- A second stage can filter false alarms.
- The positive class is rare and valuable to find.
- Screening, fraud, safety, failure prediction.

THE TIEBREAKER

They trade against each other directly. Raising one lowers the other, because both move with the same threshold.

F1, and when to weight it

- **F1 is the harmonic mean** of precision and recall. It punishes imbalance between them.
- Precision 1.0 with recall 0.1 gives an F1 of about 0.18, not 0.55. That is the point.
- Use it when you need **one number** and both errors matter roughly equally.
- **F-beta** weights one side. Beta above 1 favours recall; below 1 favours precision.
- F2 is the common recall-weighted choice. F0.5 is the precision-weighted one.

ON THE EXAM

The harmonic mean is the whole design. An arithmetic mean would let a model hide a terrible recall behind a perfect precision.

When accuracy is actually fine

- **Accuracy = (TP + TN) / everything.** The proportion of predictions that were correct.
- It is fine when classes are **roughly balanced** and both errors cost about the same.
- It is misleading the moment either of those stops being true.
- The trap is that it is the most intuitive metric, so it is what stakeholders ask for.
- Balanced accuracy averages recall across classes, which repairs part of the problem.

> ON THE EXAM
>
> Accuracy is not wrong. It is wrong by default on the problems people actually build models for, which are usually imbalanced.

A classifier outputs a **probability**. The threshold turns it into a decision, and you choose it.

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Threshold selection

- **Lower the threshold** and you flag more: recall rises, precision falls.
- **Raise the threshold** and you flag less: precision rises, recall falls.
- Choose it from the **business cost** of each error, not from a default.
- It is chosen **after** training, on validation data, and can be changed without retraining.
- It must be tuned on validation data at the **real class distribution**, not a resampled one.

> ON THE EXAM
>
> This is the cheapest lever in the whole of Domain 2. A model that misses too much often needs a threshold change, not retraining.

Probability calibration

- A **calibrated** model's 0.7 means the event happens about 70 percent of the time.
- Many models are **not** calibrated. Boosted trees often push scores toward 0 and 1.
- It matters whenever the **score itself is used**, not just the ranking above a threshold.
- Expected-value decisions, risk pricing and human review all consume the probability directly.
- Fix with Platt scaling or isotonic regression, fitted on held-out data.

> ON THE EXAM
>
> AUC can be perfect on an uncalibrated model, because ranking is fine. Calibration is separate from discrimination.

Cost matrices and expected cost

- Assign a **cost to each cell** of the confusion matrix, in money or in harm.
- Then choose the threshold that **minimises total expected cost**, not the one maximising a metric.
- A missed fraud costs the transaction value. A false alarm costs support time and goodwill.
- This turns a modelling argument into an arithmetic one, which stakeholders can check.
- It requires someone to state the costs, which is often the hardest part of the exercise.

> ON THE EXAM
>
> F1 assumes the two errors are equally bad. A cost matrix is what you use when you know they are not, and by how much.

ROC and AUC

- **The ROC curve** plots true positive rate against false positive rate, across every threshold.
- **AUC** is the area under it: the probability the model ranks a random positive above a random negative.
- **0.5 is random guessing. 1.0 is perfect.** Below 0.5 means the model is inverted.
- It is **threshold-independent**, so it measures ranking quality rather than one operating point.
- On severely imbalanced data it looks **optimistic**, because true negatives dominate the false positive rate.

> ON THE EXAM
>
> AUC answers 'is the ranking any good'. It does not answer 'is this model useful at the threshold I will actually deploy'.

ROC-AUC or PR-AUC?

ROC-AUC

TPR against FPR

- Uses true negatives in its denominator.
- Stable as class balance changes.
- Good when classes are roughly balanced.
- Optimistic when negatives vastly outnumber positives.

PR-AUC

Precision against recall

- Ignores true negatives entirely.
- Reflects performance on the rare class.
- Better for severe imbalance.
- Baseline equals the positive class rate, not 0.5.

THE TIEBREAKER

With 0.2 percent positives, a huge pile of true negatives flatters ROC-AUC. Precision-recall AUC does not have that pile to hide behind.

Regression metrics

METRIC	WHAT IT DOES
MAE	Mean absolute error. Average size of error, in the original units
MSE	Mean squared error. Squares errors, so large ones dominate
RMSE	Root of MSE. Back in original units, still punishing large errors
MAPE	Mean absolute percentage error. Relative, so scale-free
R squared	Proportion of variance explained. 1.0 is perfect, 0 is no better than the mean

ON THE EXAM

MAE against RMSE is the real decision, and it turns on whether a few large errors matter more than many small ones.

Metrics for time series

METRIC	WHAT IT ADDS
MAPE	Percentage error. Breaks down when actuals are near zero
WAPE	Weighted absolute percentage error. Handles zeros and low volume better
Quantile loss	Scores a probabilistic forecast at a quantile. What DeepAR produces
RMSE	Still applies, and still punishes large misses hardest
Bias	Are you systematically over or under forecasting? Direction, not magnitude

ON THE EXAM

Bias is the one people omit. A forecast wrong by 5 percent in both directions differs from one always 5 percent low.

When order is what matters

- Recommendation does not predict a class. It produces a **ranked list**.
- **Precision at k** and **recall at k** measure the top k results only.
- **NDCG** rewards putting the most relevant items **highest**, not merely in the list.
- **MAP** averages precision across the ranked positions where relevant items appear.
- Module 09 used the same family for retrieval quality. Same metrics, different application.

> ON THE EXAM
>
> A recommender that puts the right item at position 40 has technically found it and practically failed.

MAE or RMSE?

MAE, WHEN

All errors count equally

- Being wrong by 10 is twice as bad as by 5.
- Outliers present, and should not dominate.
- You want a number stakeholders grasp.
- Robust to a few extreme values.

RMSE, WHEN

Large errors are disproportionately bad

- Being wrong by 10 is worse than twice by 5.
- One big miss causes real damage.
- Squaring lets large errors dominate.
- Sensitive to outliers, deliberately.

THE TIEBREAKER

Ask whether the harm of an error grows linearly or faster. Under-stocking by 1,000 units is usually much worse than twice under-stocking by 500.

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R squared

- The proportion of variance in the target that the model explains.
- **1.0 is perfect. 0 means no better than always predicting the mean.** It can go negative.
- It always rises when you add features, even useless ones. **Adjusted R squared** corrects for that.
- A high value does not mean the predictions are accurate enough to be useful.
- It is a comparison against the mean, not against a requirement. Those are different questions.

> ON THE EXAM
>
> R squared of 0.85 sounds excellent and tells you nothing about whether the errors are small enough for the business.

Averaging across classes

- **Macro average:** compute the metric per class, then average. Every class counts equally.
- **Micro average:** pool all predictions, then compute once. Large classes dominate.
- **Weighted average:** per-class metric, weighted by class size.
- **Macro exposes poor performance on rare classes.** Micro hides it.
- Reporting only micro on an imbalanced multiclass problem repeats the accuracy trap.

> ON THE EXAM
>
> If the rare classes are the ones you care about, macro averaging is the honest choice.

The right metric is the one that **moves when the business outcome moves.**

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Which metric

THE SITUATION	THE METRIC
Balanced classes, errors cost the same	Accuracy
Missing a positive is dangerous	Recall, or F2
A false alarm is expensive or disruptive	Precision, or F0.5
Both errors matter, one number needed	F1
Rare positive class, comparing models	PR-AUC
Ranking quality, roughly balanced	ROC-AUC
Predicting a number, outliers present	MAE
Predicting a number, big misses are costly	RMSE
Multiclass with rare classes that matter	Macro-averaged F1

ON THE EXAM

Nine rows, and every one is chosen by asking what the errors cost. That question comes before the metric, always.

Baselines

- A metric with no baseline is meaningless. Is 0.82 good? You cannot say.
- **Majority class:** always predict the most common. That is the floor for classification.
- **Predict the mean:** the floor for regression, and exactly what R squared compares against.
- **The existing process:** the rules engine or the human currently doing this job.
- That last one is the only baseline the business recognises as meaningful.

> ON THE EXAM
>
> A model that beats random and loses to the current rules engine is not an improvement, however good the metric looks.

Evaluate by segment, not only overall

- An overall metric can be excellent while one group is served badly.
- Break results down by **segment, region, device, time period** and any protected attribute.
- This is where **module 08's bias work rejoins evaluation**: post-training disparity shows up here.
- New or rare segments are usually the weakest, because they had least training data.
- Report the worst segment alongside the average. The average alone is not a safety claim.

> ON THE EXAM
>
> Simpson's paradox from module 08 applies here too: an aggregate can hide, and even invert, what happens in subgroups.

Sample size and meaningful differences

- Two models scoring 0.834 and 0.829 may be identical. The gap can be noise.
- The **smaller the test set**, the larger a difference must be before it means anything.
- **Cross-validation** gives a spread rather than a point, which shows the uncertainty.
- That is why the evaluation report holds a **standard_deviation** next to each value.
- Rare positive classes make this worse: a handful of positives in test is a noisy estimate.

> ON THE EXAM
>
> Picking the model with the higher third decimal place is a common and expensive way to choose randomly.

Validation and test are not the same thing

- **Validation** is used during development: early stopping, tuning, threshold selection.
- Because you made decisions with it, it is **no longer an unbiased estimate**.
- **Test** is touched once, at the end, to report expected production performance.
- Checking the test set repeatedly and adjusting turns it into a second validation set.
- Module 15's tuning used validation for exactly this reason. Tuning on test would report a fiction.

> ON THE EXAM
>
> Every decision you make using a dataset consumes a little of its honesty. Test survives only if you leave it alone.

Comparing two models honestly

- **Same test set, same preprocessing, same metric, same threshold policy.** Change one thing only.
- Comparing a model at its best threshold with another at 0.5 is not a comparison.
- Compare **quality, latency and cost together**. The best model is often not the one you ship.
- Include the **current production model** as a candidate. Beating last month is the actual bar.
- Shadow testing, in module 24, is how you compare on live traffic without risk.

> ON THE EXAM
>
> Most model comparisons in practice are unfair by accident, because something other than the model changed between runs.

Evaluation as a Processing job

- Evaluation runs as a **Processing job step**, loading the model and scoring the test set.
- It writes an **evaluation report to S3**, conventionally evaluation.json.
- Using a Processing job means the **same libraries as training**, from module 04. No skew in the metric itself.
- For a large test set, batch transform generates the predictions and Processing computes the metrics.
- Module 28 puts this in a pipeline. Here it is just the mechanism.

> ON THE EXAM
>
> Evaluation is code, and code that computes your headline number deserves the same rigour as the training script.

What evaluation.json contains

TOP-LEVEL KEY	HOLDS
`regression_metrics`	mae, mse, rmse, r2 and similar
`binary_classification_metrics`	accuracy, precision, recall, f1, auc, confusion_matrix
`multiclass_classification_metrics`	the same, averaged across classes
`Each metric`	An object: value, and optionally standard_deviation

ON THE EXAM

The structure matters because downstream steps read specific paths out of it. A ConditionStep reads one value from this file.

Gating registration with a ConditionStep

- A **ConditionStep** reads a metric from the evaluation report and branches on it.
- If the model does not clear the bar, it is **never registered**, so it cannot be deployed.
- This is how a quality bar becomes automatic rather than a habit somebody might forget.
- The threshold is a pipeline parameter, so it can be raised without editing the pipeline.
- Gate on the **right** metric. Gating on training accuracy guarantees everything passes.

> ON THE EXAM
>
> An automated retraining pipeline with no condition step will eventually deploy a worse model, quietly, on a Tuesday.

Reporting a result honestly

REPORT	RATHER THAN
The metric, and what it means in cases	A single number with no interpretation
Performance against the current process	Performance against random
The worst segment, not just the average	One aggregate figure
The operating threshold you will deploy	A threshold-independent score alone
What the model gets wrong, and how often	Only what it gets right

ON THE EXAM

'Recall of 0.86 means we catch about 86 of every 100 fraud cases, and miss 14' is a sentence a stakeholder can act on.

Model metrics in the registry

- **RegisterModel** takes model_metrics, so the numbers travel with the model version.
- It also takes **drift_check_baselines**, which Model Monitor later checks against.
- **Five baselines** can be registered: data quality, model quality, data bias, model bias, explainability.
- QualityCheck produces two files, statistics and constraints. ClarifyCheck produces one, constraints.
- Model Monitor is closed to new customers, from module 08. The baseline concept remains examinable.

> ON THE EXAM
>
> This is the link between evaluation and monitoring. What you measured at training time becomes what production is judged against.

Explainability is part of evaluation

- A metric says **how well**. Explainability says **why**, and the two answer different questions.
- A model can score well for a reason that will not survive contact with production.
- Feature importance dominated by one column is a **leakage check**, from module 05.
- Regulated decisions often require per-prediction explanation, from module 12.
- Module 19 covers SHAP, Clarify and experiment tracking properly.

ON THE EXAM

Evaluation asks two questions, not one: is it good enough, and is it good for defensible reasons?

Symptoms, and what they usually mean

SYMPTOM	USUAL CAUSE
99 percent accurate, catches nothing	Imbalanced classes. Accuracy is the wrong metric
Excellent AUC, useless in production	Ranking is fine; the deployed threshold is wrong
Great in validation, worse in test	Validation was used for too many decisions
High R squared, business unimpressed	Variance explained is not the same as errors being small enough
Multiclass looks fine, rare classes fail	Micro averaging hiding it. Use macro
Fraud recall great in test, poor live	Test set was resampled. Module 08
Every retrained model passes the gate	Gating on a training metric, not a held-out one

ON THE EXAM

Row two is worth acting on: a good model with a bad threshold is a configuration fix, not a retraining project.

Choose the metric from the **cost of being wrong**, before you look at any model.

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> A hospital builds a model to flag chest X-rays that may show early-stage disease. Flagged images go to a radiologist for review; unflagged images are not reviewed again.
>
> Radiologist capacity allows roughly three times the current flag rate. Missing a case delays diagnosis by months.

Which metric should the model be optimised for?

A Precision, to avoid wasting radiologist time on false alarms

B Recall, because an unflagged image is never reviewed again

C Accuracy, since both kinds of error affect patients

D ROC-AUC, because it is threshold-independent

◆ PAUSE AND DECIDE

Which metric should the model be optimised for?

A. Precision, to avoid wasting radiologist time on false alarms

Would be right if review capacity were the binding constraint. The scenario says capacity allows three times the current flag rate, so it is not.

B. Recall, because an unflagged image is never reviewed again

A false negative is unrecoverable: no second stage catches it. A false positive costs review time, which the scenario says is available.

C. Accuracy, since both kinds of error affect patients

Early-stage disease is rare, so accuracy is dominated by true negatives. A model flagging nothing would score very well.

D. ROC-AUC, because it is threshold-independent

Useful for comparing models during development. It does not express which error this deployment can least afford.

THE SIGNAL

The deciding sentence is 'unflagged images are not reviewed again'. That makes a false negative permanent and a false positive merely costly.

> A payments team reports their new fraud model at 99.81 percent accuracy and asks to deploy. Fraud is 0.19 percent of transactions.
>
> Reviewing the confusion matrix, the model flags almost nothing. The team wants to know which metric they should have reported instead.

What should they report?

A. Balanced accuracy, which averages recall across both classes

B. ROC-AUC, since it is insensitive to the chosen threshold

C. R squared, to show how much variance the model explains

D. PR-AUC together with recall at the deployed threshold

◆ PAUSE AND DECIDE

What should they report?

A. Balanced accuracy, which averages recall across both classes

A genuine improvement on accuracy, and it still reports one summary number with no view of precision or the operating point.

B. ROC-AUC, since it is insensitive to the chosen threshold

With 0.19 percent positives, the vast pile of true negatives flatters the false positive rate. It can look strong while the model catches almost nothing.

C. R squared, to show how much variance the model explains

A regression metric. This is binary classification, so it does not apply at all.

D. PR-AUC together with recall at the deployed threshold

PR-AUC ignores true negatives, so severe imbalance cannot flatter it. Recall at the deployed threshold states what the model actually catches in production.

THE SIGNAL

Two numbers, two jobs: PR-AUC compares models honestly under imbalance, and recall at threshold describes the model you are actually shipping.

A retailer forecasts daily demand per store. Being over by a few units means minor holding cost. Being under by a large amount means empty shelves, lost sales and a customer who shops elsewhere next week.

Two candidate models are close. Model A has slightly lower average error but occasionally misses badly. Model B is consistently a little wrong.

Which metric should decide between them?

A MAE, because it reports the average error in units the business understands

B MAPE, because it is scale-free across stores of different sizes

C RMSE, because squaring makes occasional large errors dominate

D R squared, because it shows which model explains more variance

◆ PAUSE AND DECIDE

Which metric should decide between them?

A MAE, because it reports the average error in units the business understands

Treats all errors as equally bad, so it would favour Model A. The scenario states large misses are disproportionately damaging.

B MAPE, because it is scale-free across stores of different sizes

A real benefit for comparing stores, and it still weights errors linearly. It does not express the harm of one big miss.

C RMSE, because squaring makes occasional large errors dominate

The harm grows faster than linearly, and squaring matches that. It penalises Model A's occasional bad misses, favouring the consistent model.

D R squared, because it shows which model explains more variance

Compares each model against predicting the mean. It does not express the asymmetric cost of the errors being made.

THE SIGNAL

The scenario describes non-linear harm, and RMSE is the non-linear metric. Match the shape of the metric to the shape of the damage.

> A team automates monthly retraining with a SageMaker Pipeline. A ConditionStep gates model registration on accuracy above 0.90, read from the evaluation report.
>
> Every model passes. Reviewing the pipeline, the evaluation Processing job scores the model on the same dataset it was trained on.

What should be changed?

A. Evaluate on a held-out test set, and gate on a metric suited to the problem

B. Raise the accuracy threshold from 0.90 to 0.98 so the gate is meaningful

C. Remove the ConditionStep and review each model manually before registration

D. Add a second ConditionStep checking training loss as well

◆ PAUSE AND DECIDE

What should be changed?

A Evaluate on a held-out test set, and gate on a metric suited to the problem

The gate measures memorisation, not generalisation. A held-out set makes the number meaningful, and the right metric makes the bar meaningful.

B Raise the accuracy threshold from 0.90 to 0.98 so the gate is meaningful

A model evaluated on its own training data can clear 0.98 while generalising badly. Raising a broken gate does not repair it.

C Remove the ConditionStep and review each model manually before registration

Discards automation because it was configured wrongly, and reintroduces the human step that automation existed to remove.

D Add a second ConditionStep checking training loss as well

Another training-data measure. Two gates both looking at training performance are no better than one.

THE SIGNAL

Note that the fix has two halves. Held-out data makes the measurement honest; the right metric makes the threshold mean something.

What to take from this module

- **Ask what each error costs** before choosing a metric. Everything follows from that.
- **Accuracy is misleading on imbalanced data.** Precision, recall, F1 or PR-AUC instead.
- **The threshold is a free lever**, chosen after training, from business cost.
- **RMSE punishes large errors. MAE does not.** Match the metric to the shape of the harm.
- **Evaluate on held-out data**, and gate registration on that metric, not a training one.

> ON THE EXAM
>
> Module 18 asks the same question about foundation models, where there is often no single correct answer to compare against.